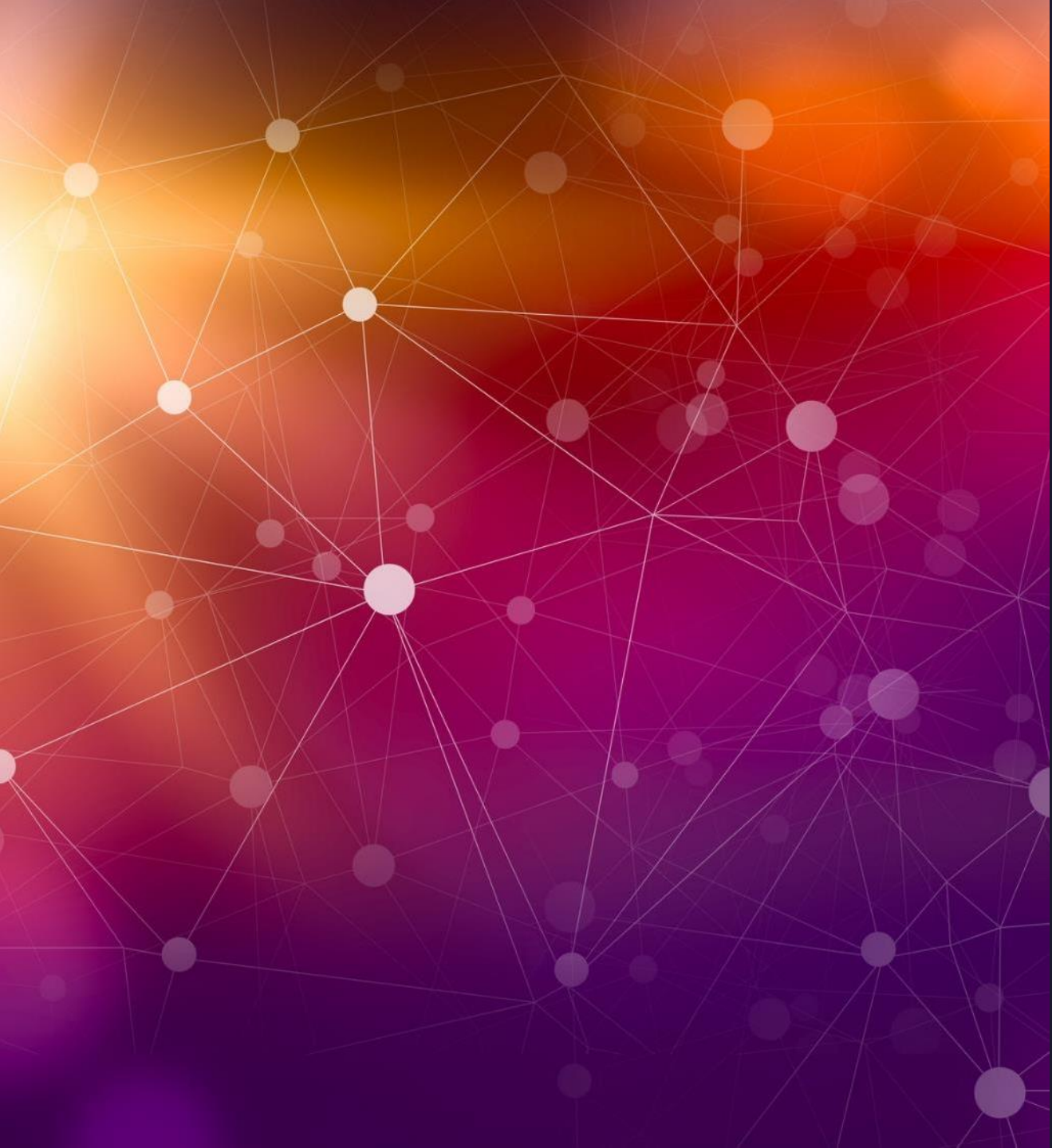




Computational Linear Algebra for Large Scale Problems

A.Y. 2025/2026

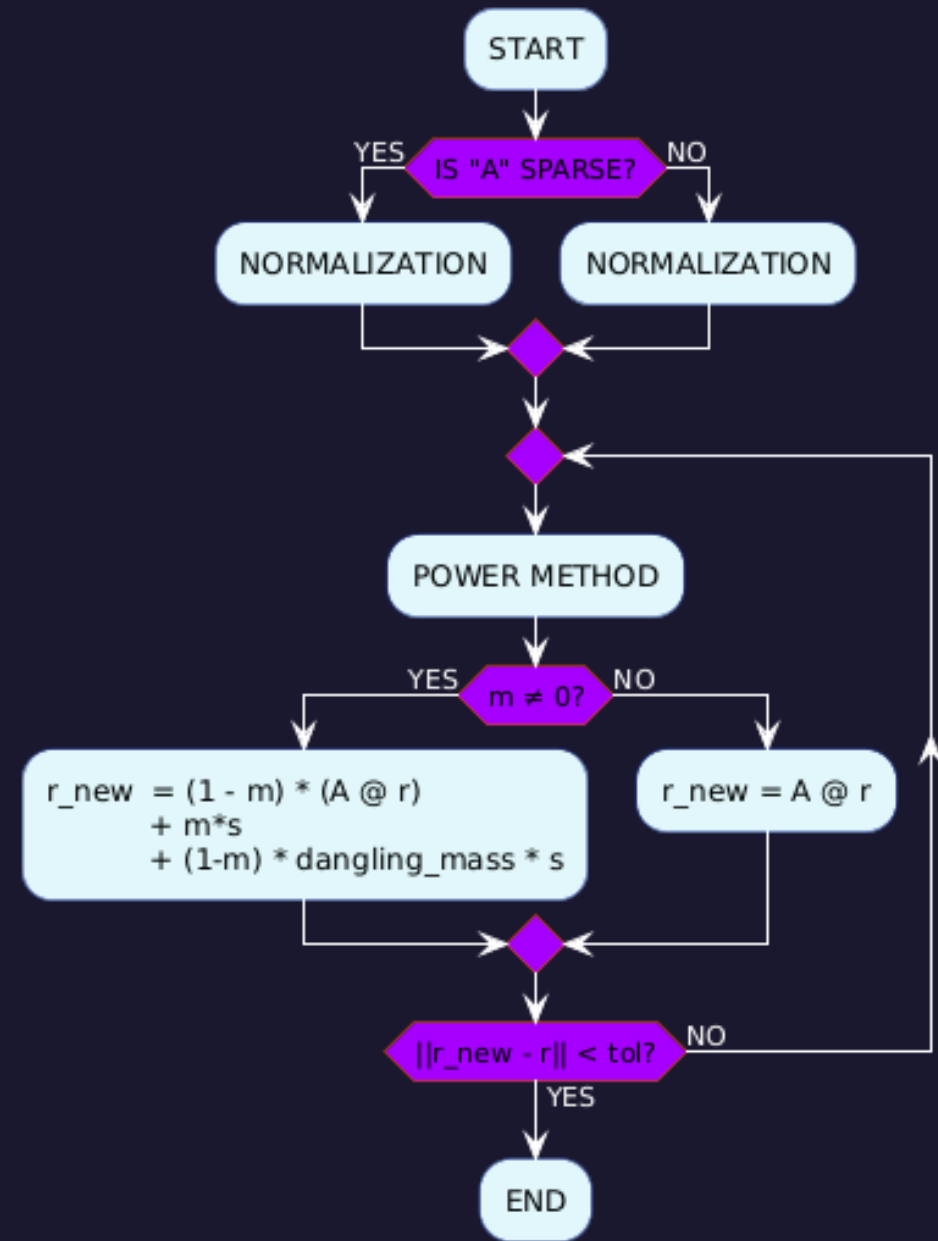
Homework on PageRank and Spectral Clustering



PageRank

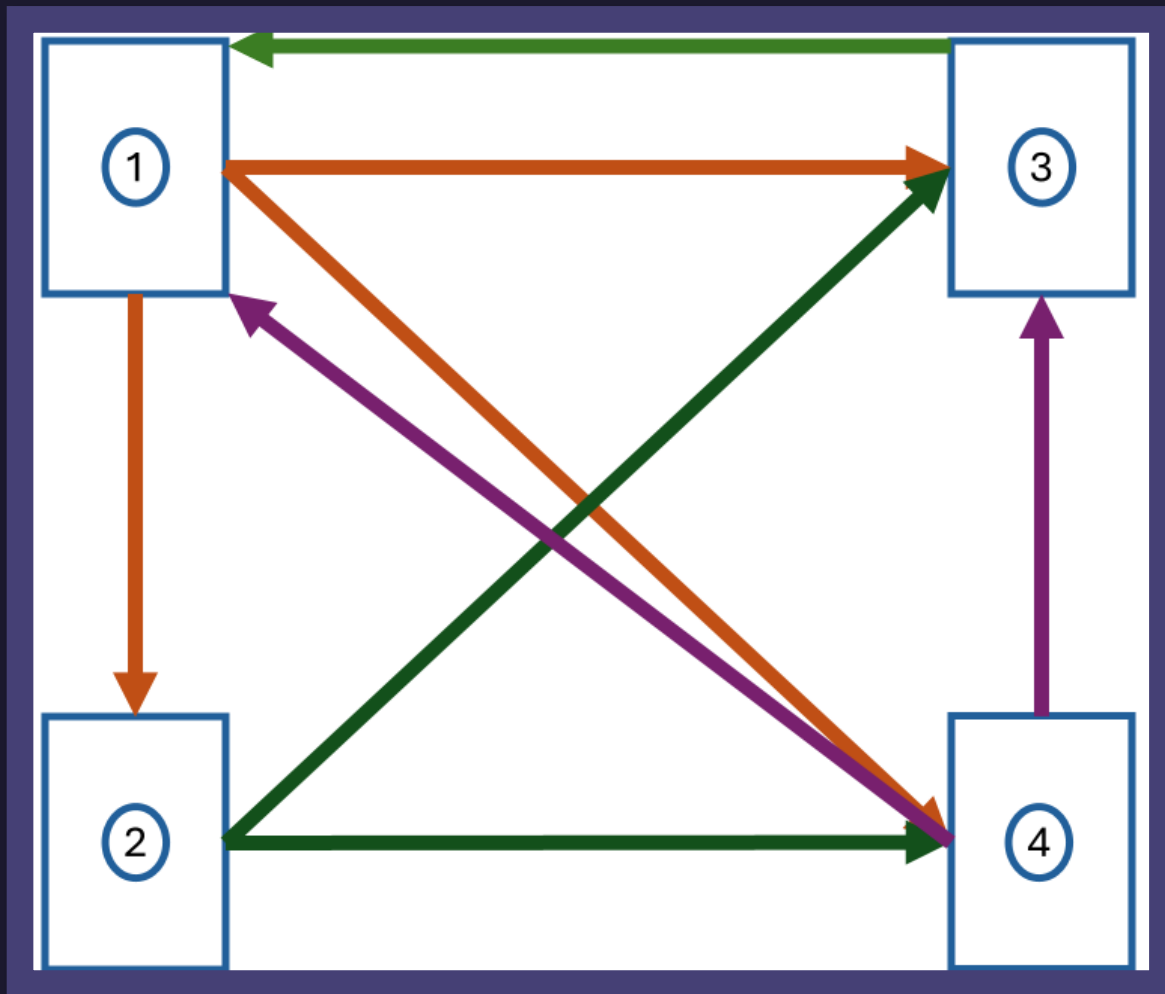
pagerank workflow

- Inputs: A , m , tol , max_iter
- Computes eigenvector associated to greatest eigenvalue
- Converges because A is made column stochastic



pagerank on small webs

Web 1



pagerank $m=0$

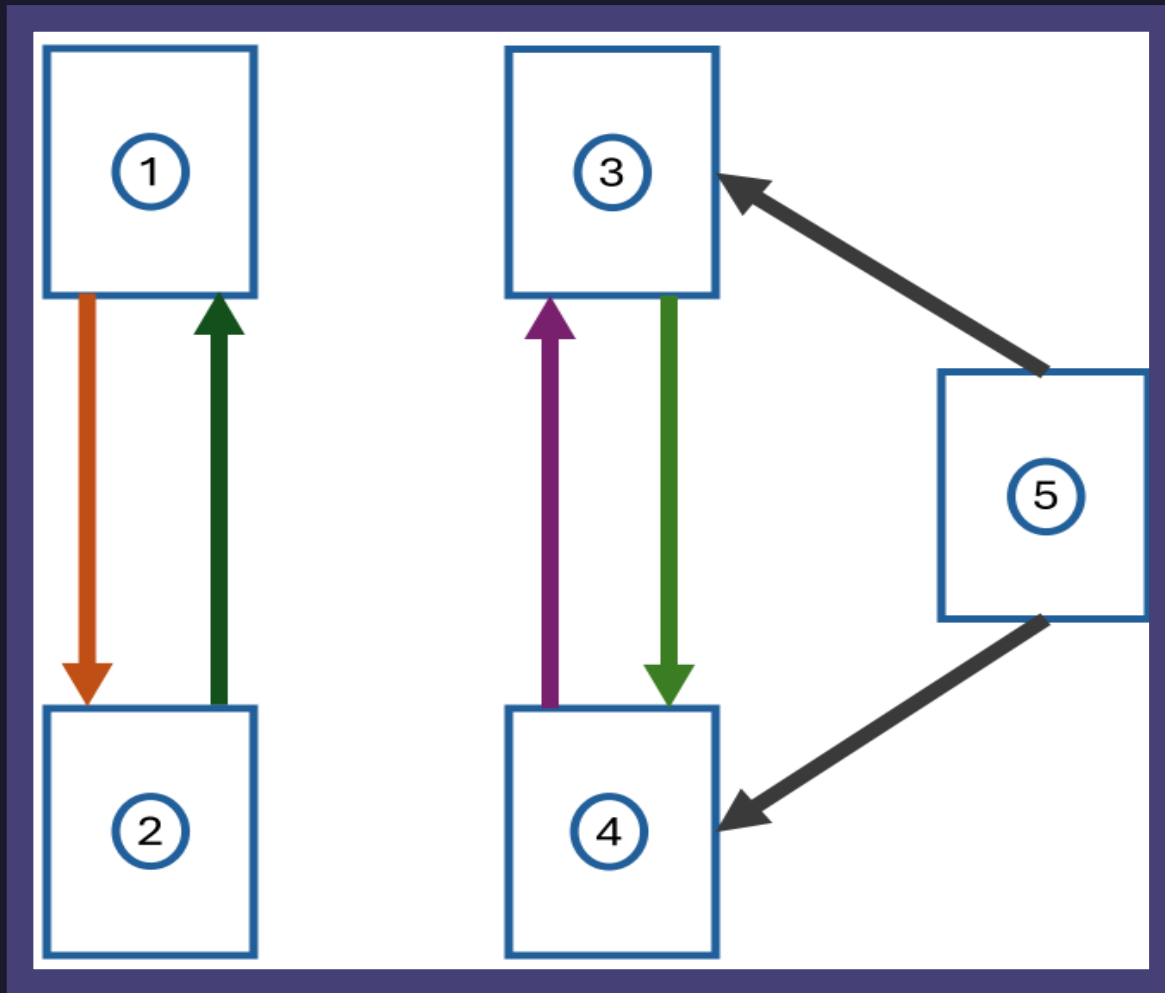
Page	Score
1	0.387
2	0.128
3	0.290
4	0.194

pagerank $m=0.15$

Page	Score
1	0.368
2	0.142
3	0.288
4	0.202

pagerank on small webs

Web 2



pagerank $m=0$

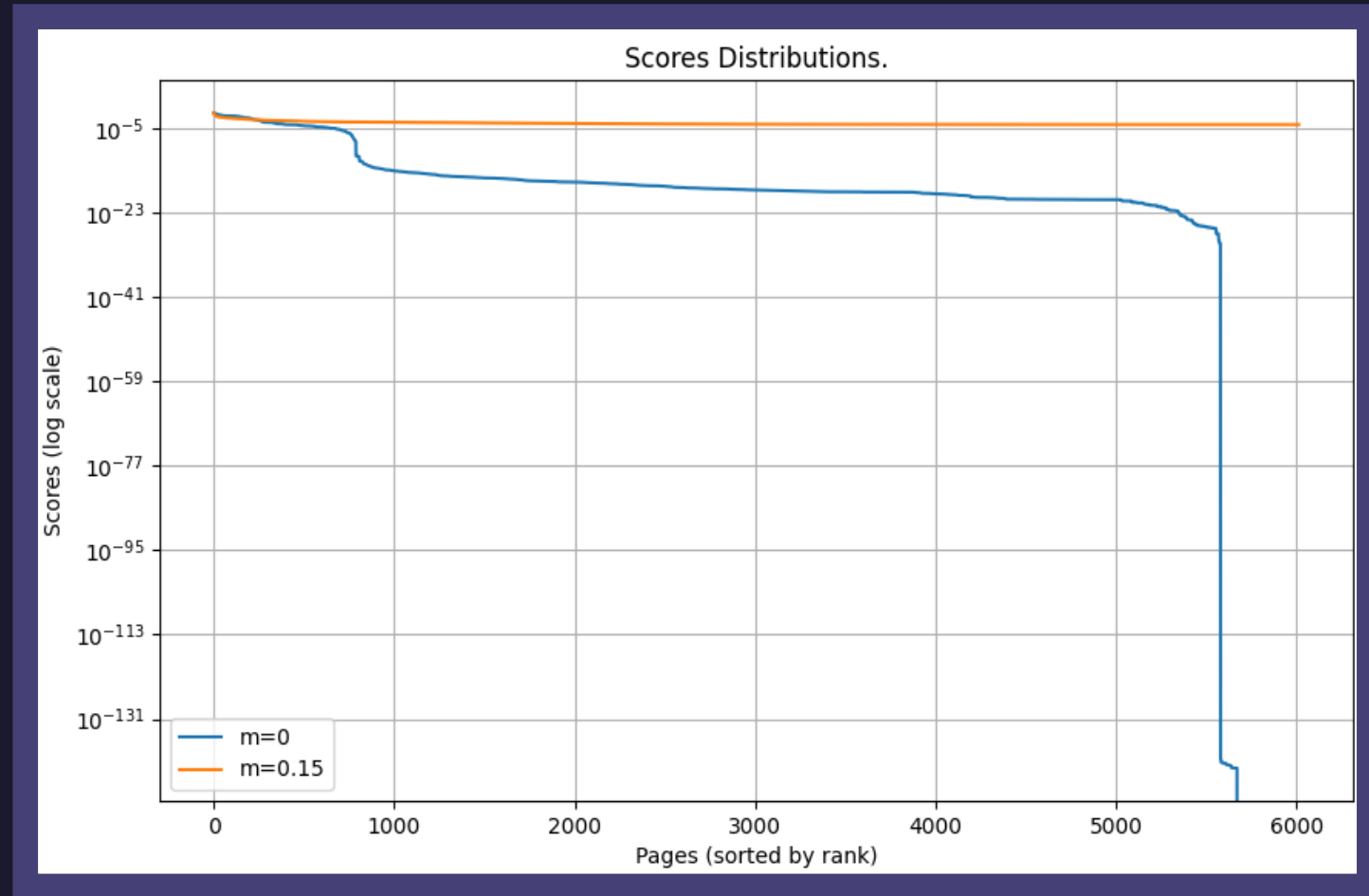
Page	Score
1	0.2
2	0.2
3	0.3
4	0.3
5	0

pagerank $m=0.15$

Page	Score
1	0.2
2	0.2
3	0.285
4	0.285
5	0.03

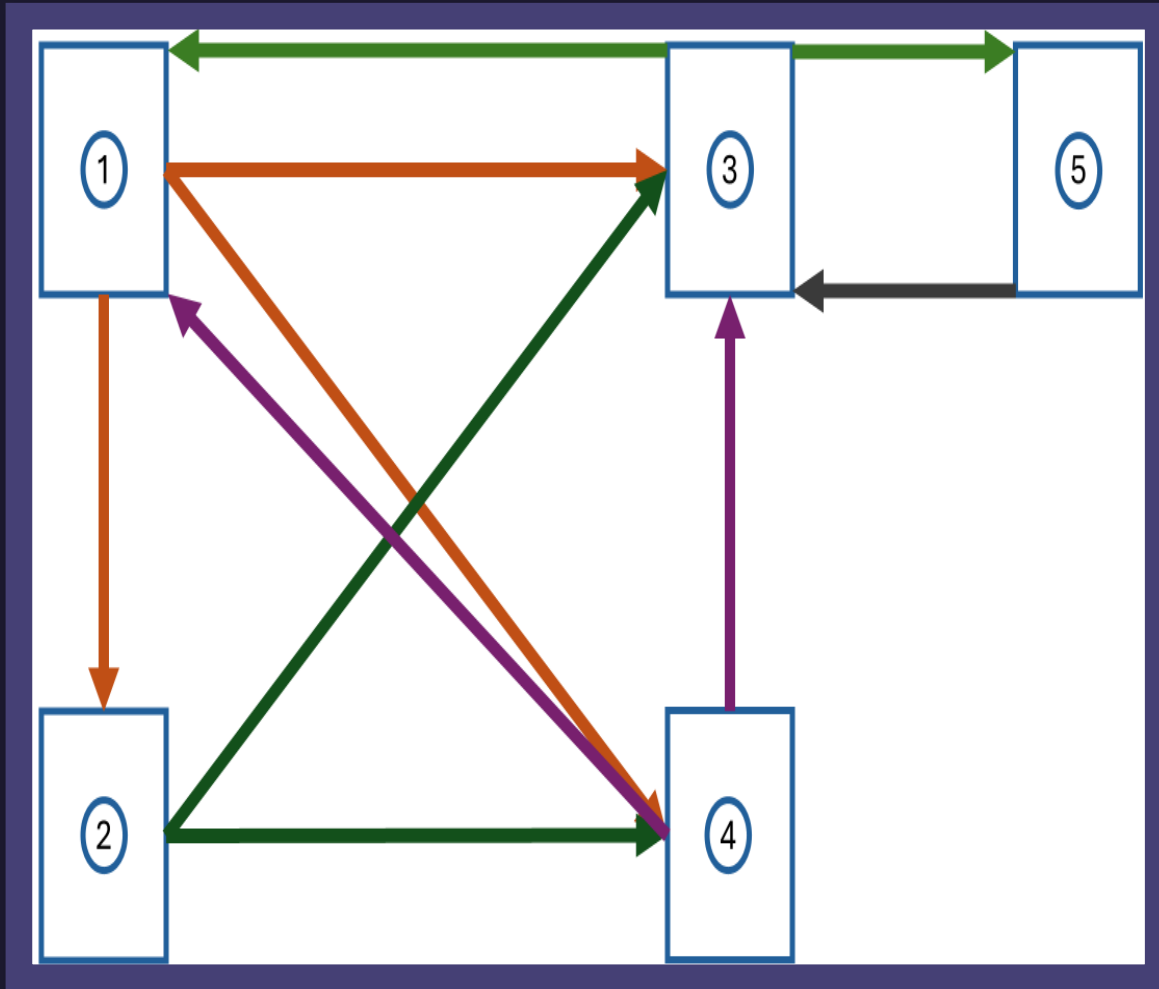
pagerank on a dataset

- Dataset of 6012 pages
- Great number of dangling nodes
- Sparsely connected pages



Exercise 11

Adding a fifth page with reciprocal link

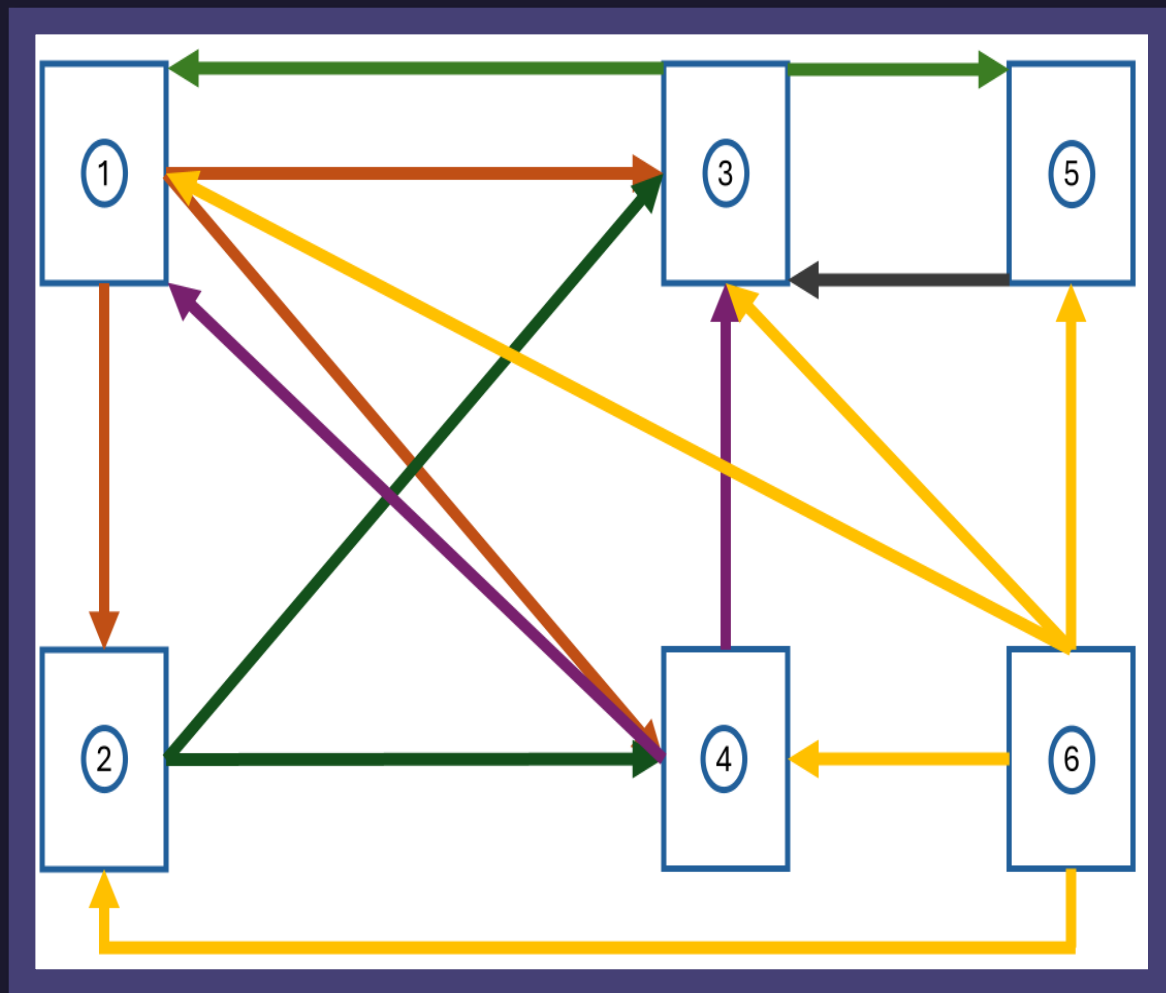


pagerank $m=0.15$

Page	Score
1	0.237
2	0.097
3	0.349
4	0.138
5	0.178

Exercise 12

Adding a sixth page that links to everything



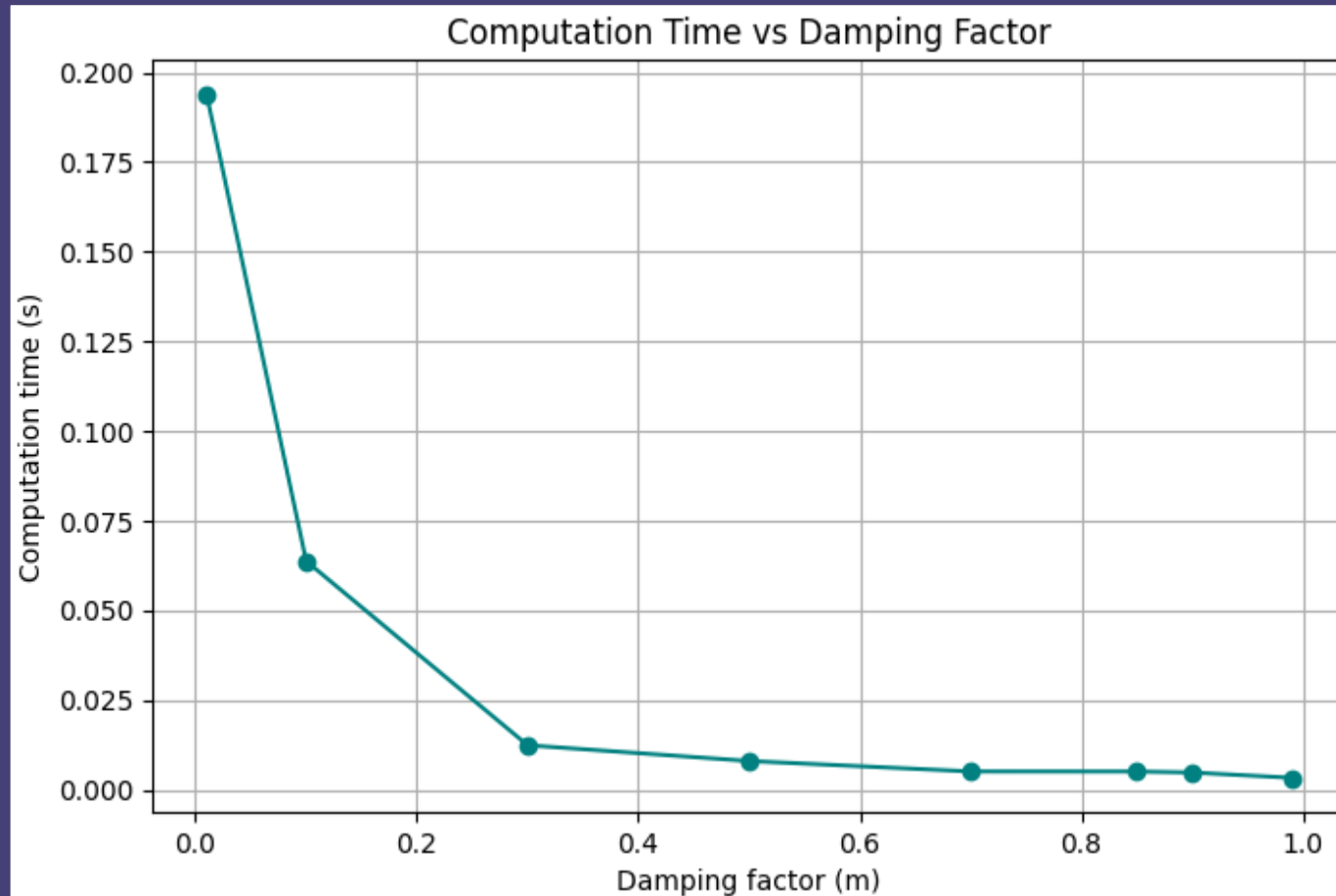
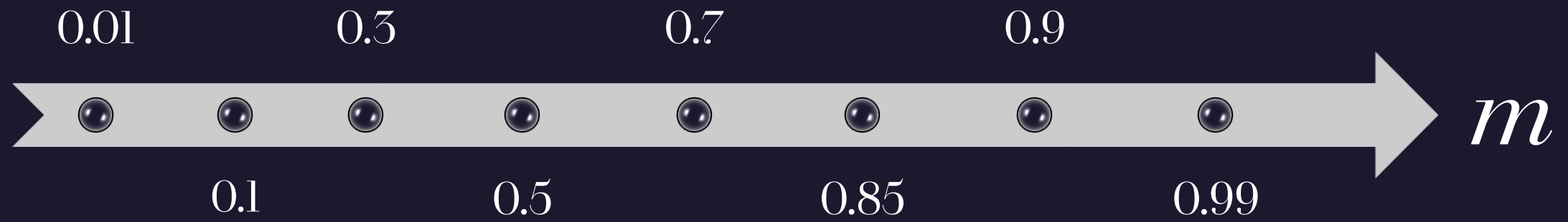
pagerank $m=0$

Page	Score
1	0.245
2	0.082
3	0.367
4	0.122
5	0.184
6	0

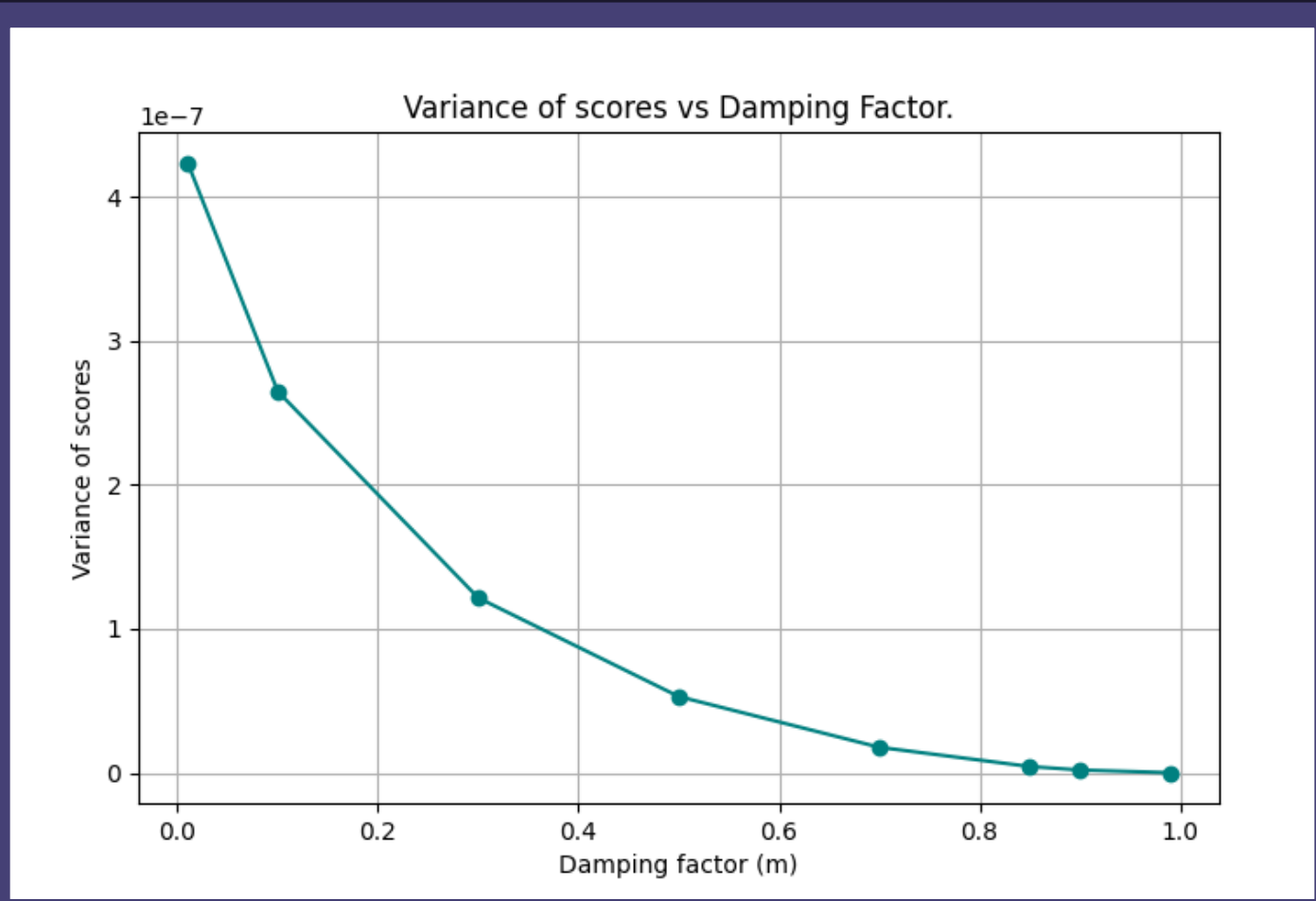
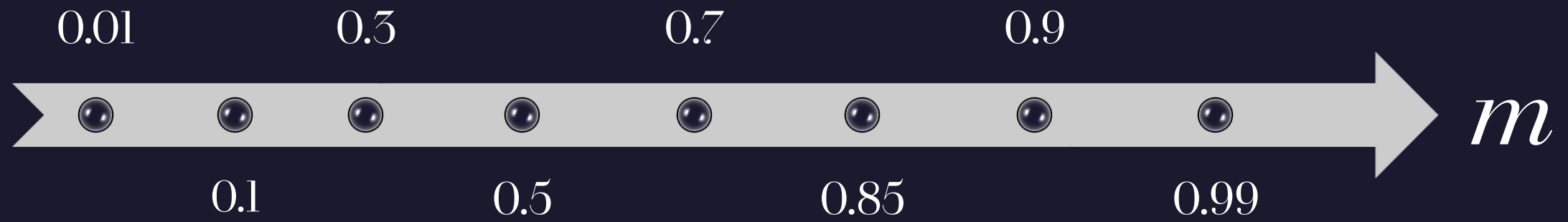
pagerank $m=0.15$

Page	Score
1	0.231
2	0.095
3	0.340
4	0.135
5	0.174
6	0.025

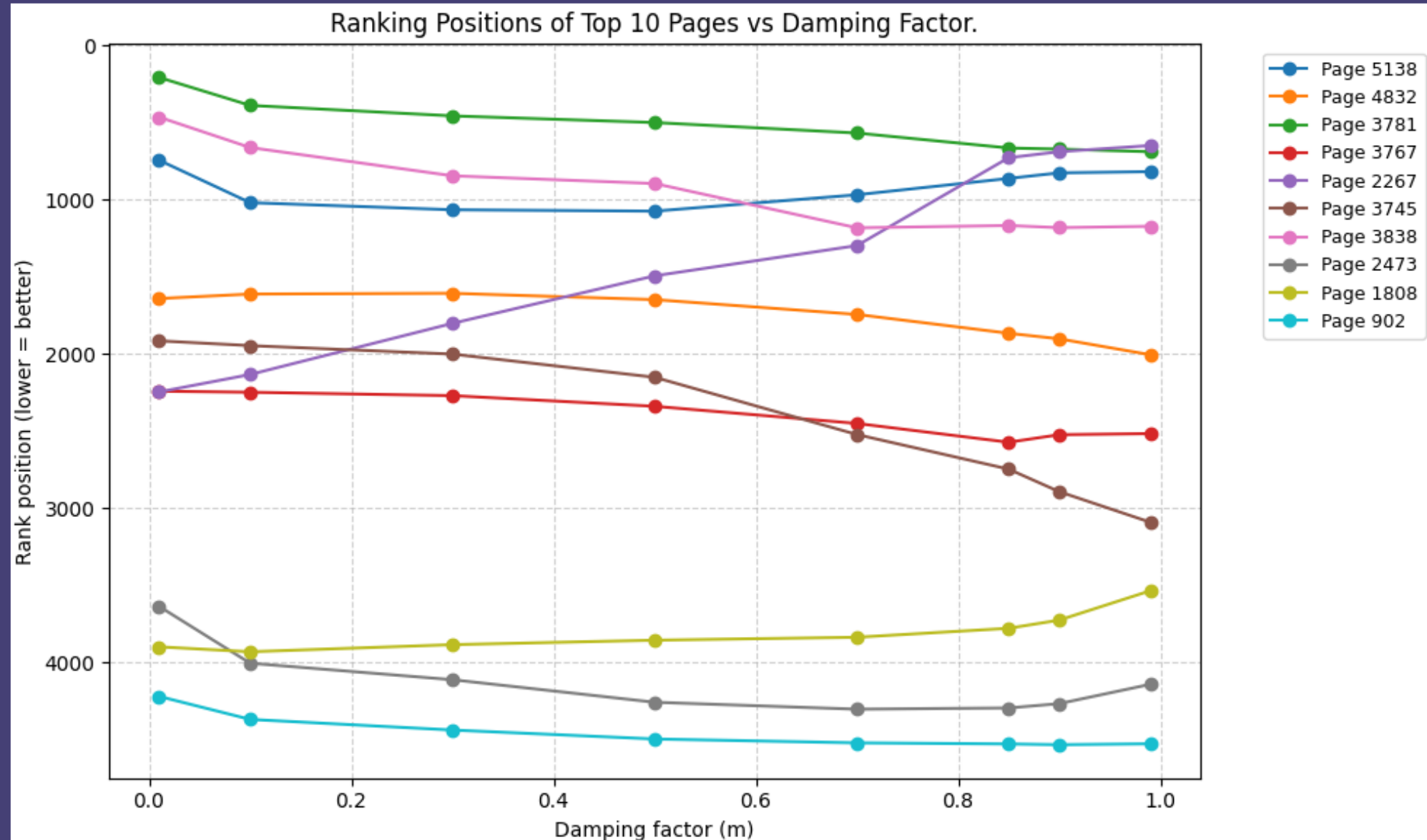
Exercise 17 - Changing m values, times



Exercise 17 - Changing m values, variances



Exercise 17 - Changing m values.

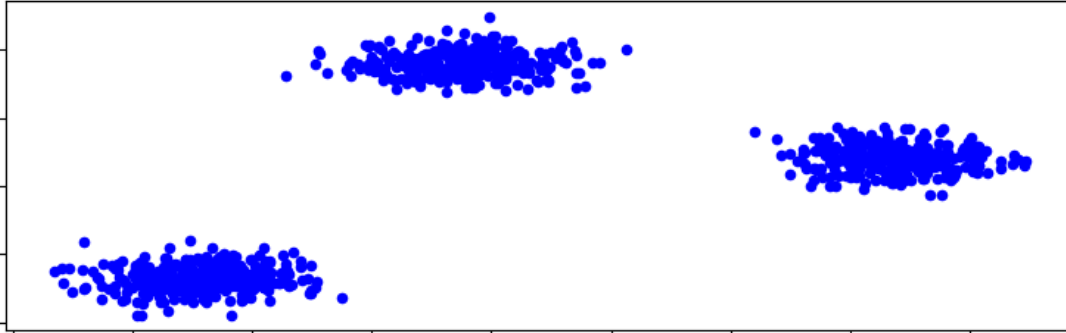




Spectral Clustering

Limitations of Kmeans

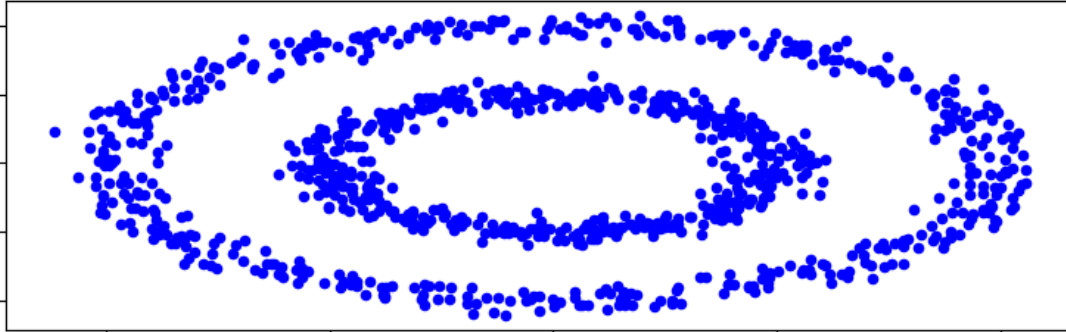
Original dataset: Gaussian Mixture Model.



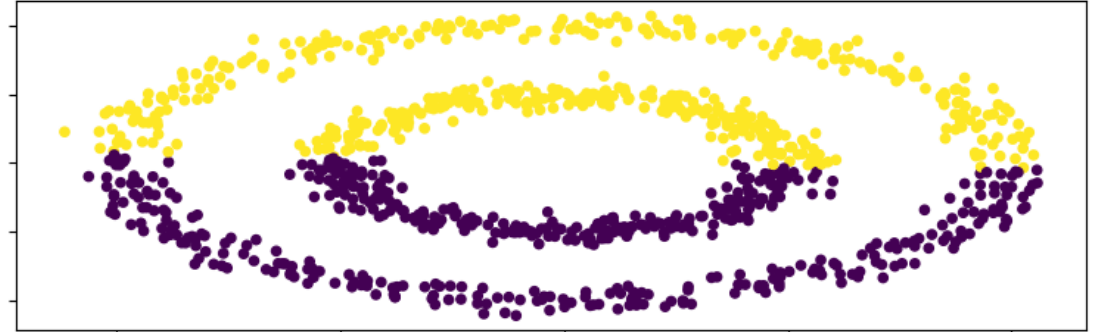
K-means.



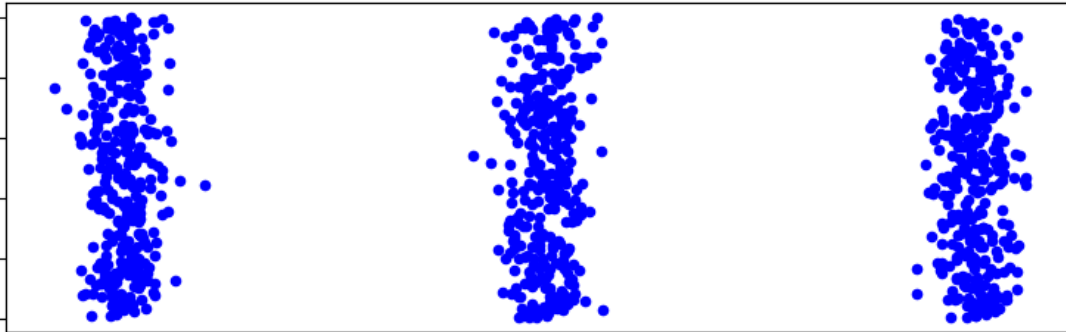
Original dataset: Concentric Circles.



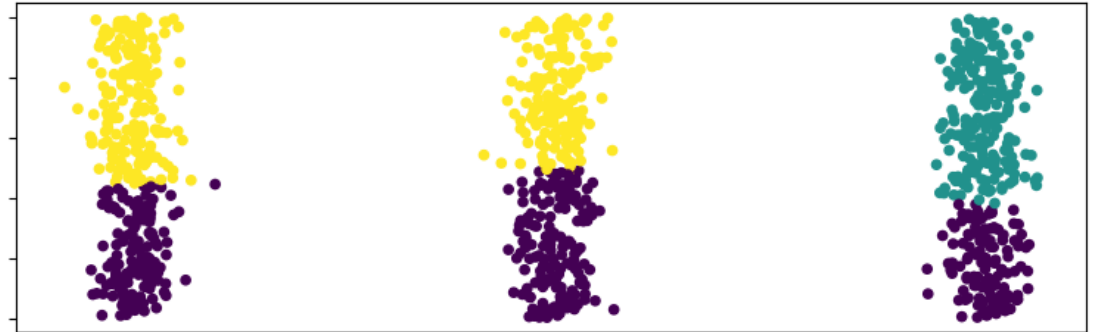
K-means.



Original dataset: Anisotropic Data.



K-means.



Spectral Clustering

Define W

Define L

$\text{eigs}(L)$

KMeans

Neighbor
Fully
connected

Normalized or
not

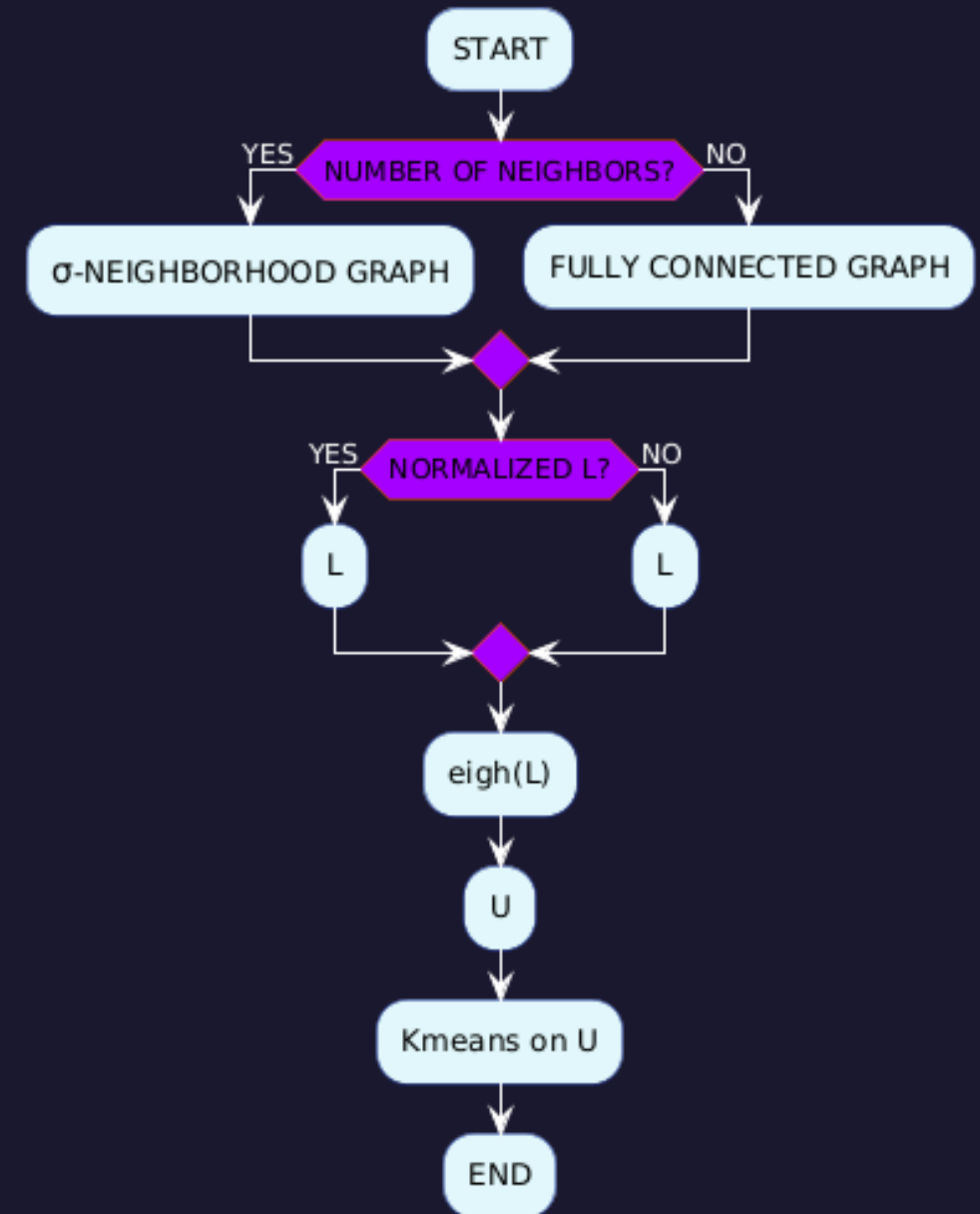
spectral_clustering workflow

- Inputs:

X , k , n° of neighbors, σ , Norm, n° of k

- Returns:

Labels, eigenvalues and eigenvectors of L

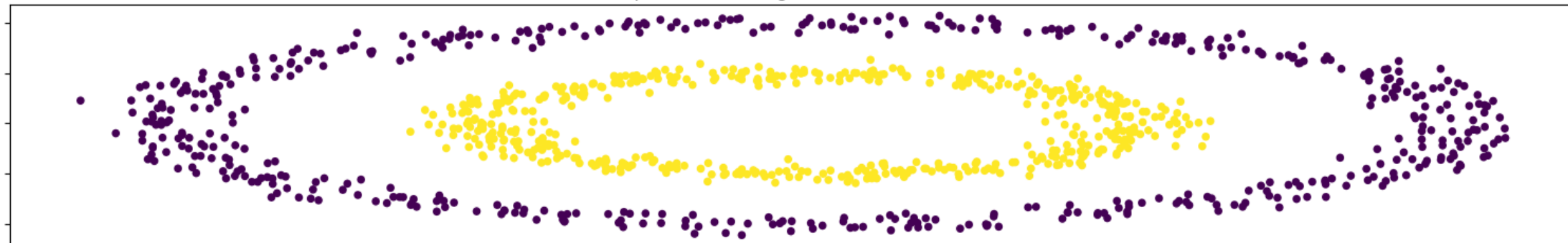


K-neighbors graph

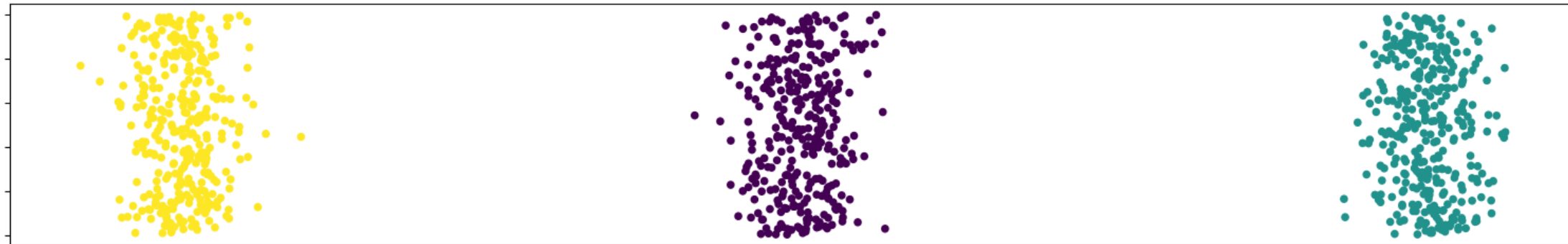
Spectral clustering - Gaussian Mixture Model



Spectral clustering - Concentric Circles



Spectral clustering - Anisotropic Data

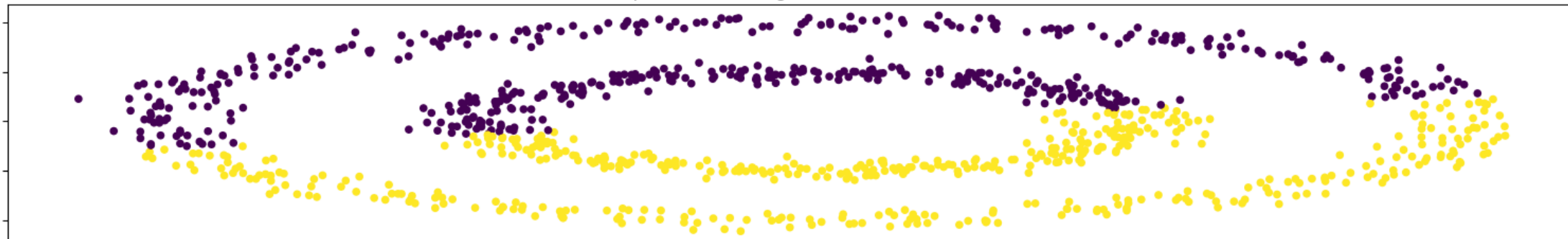


Fully connected graph

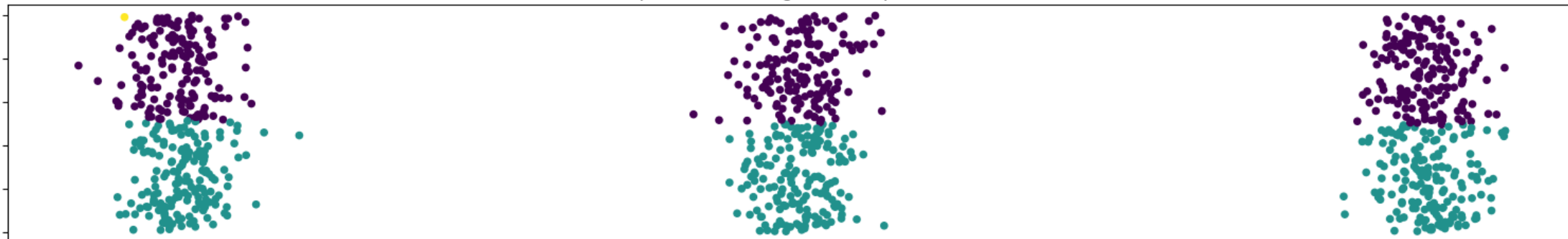
Spectral clustering - Gaussian Mixture Model



Spectral clustering - Concentric Circles



Spectral clustering - Anisotropic Data

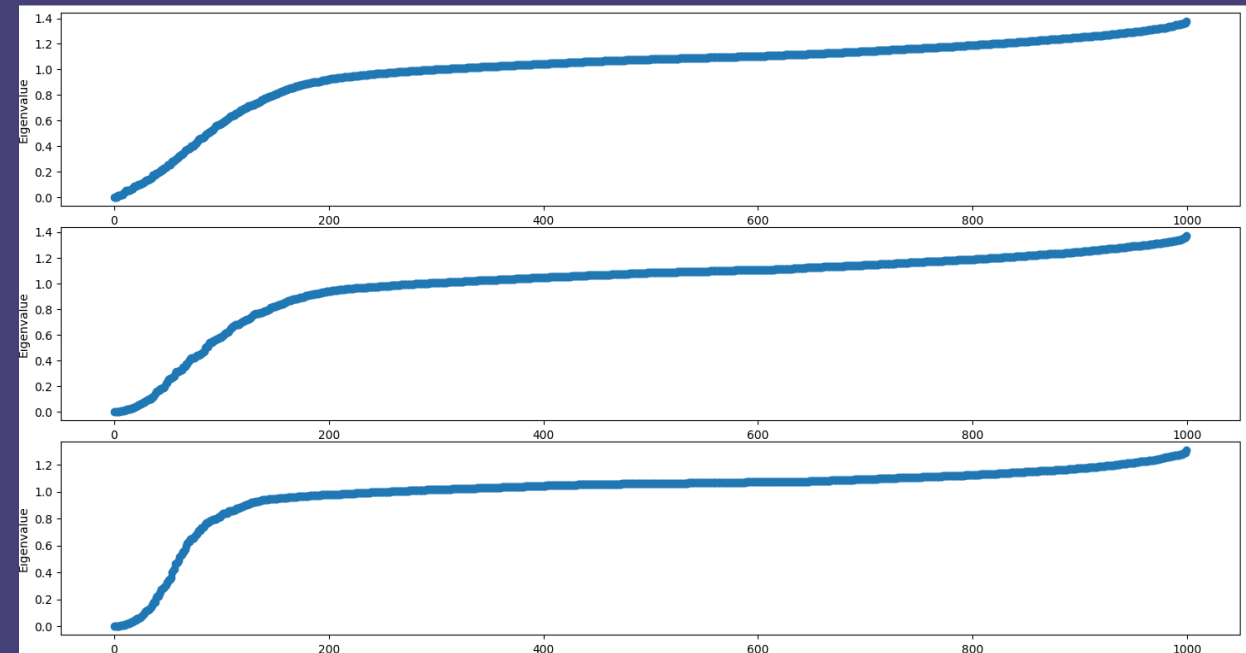
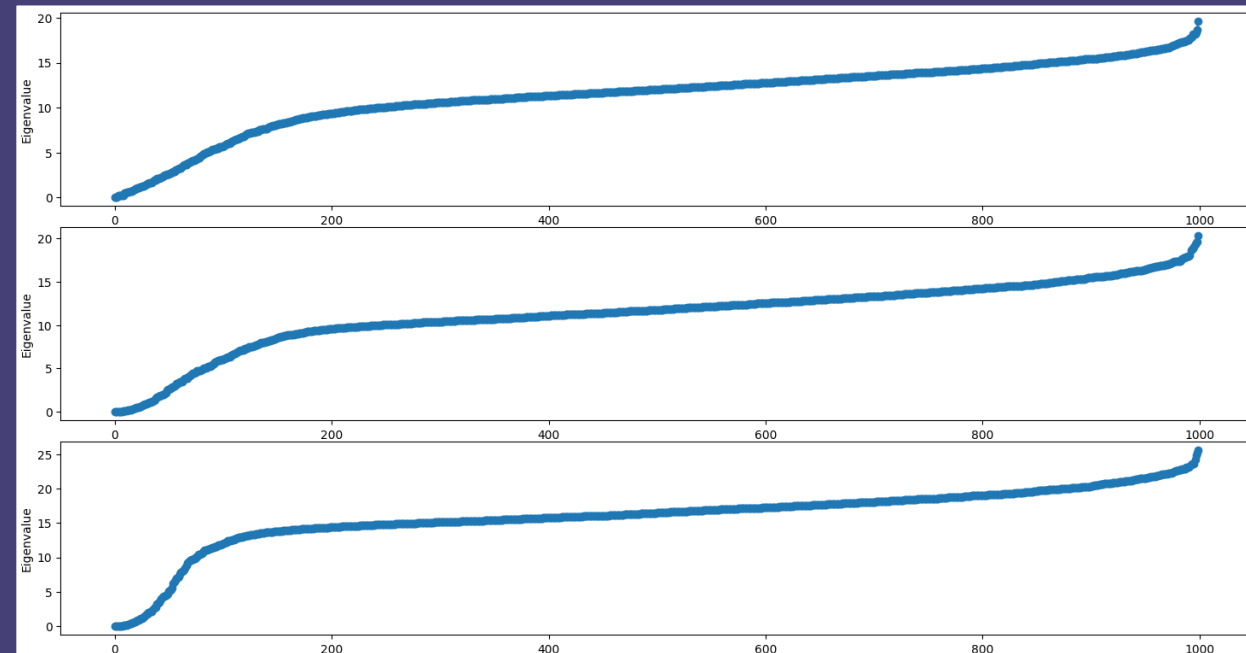


Spectra of Graph Laplacian



$$L = D - W$$

$$L = -D^{\frac{1}{2}}LD^{-\frac{1}{2}}$$



Exercise 1 – Eigenvectors of n -cycle

Adjacency matrix

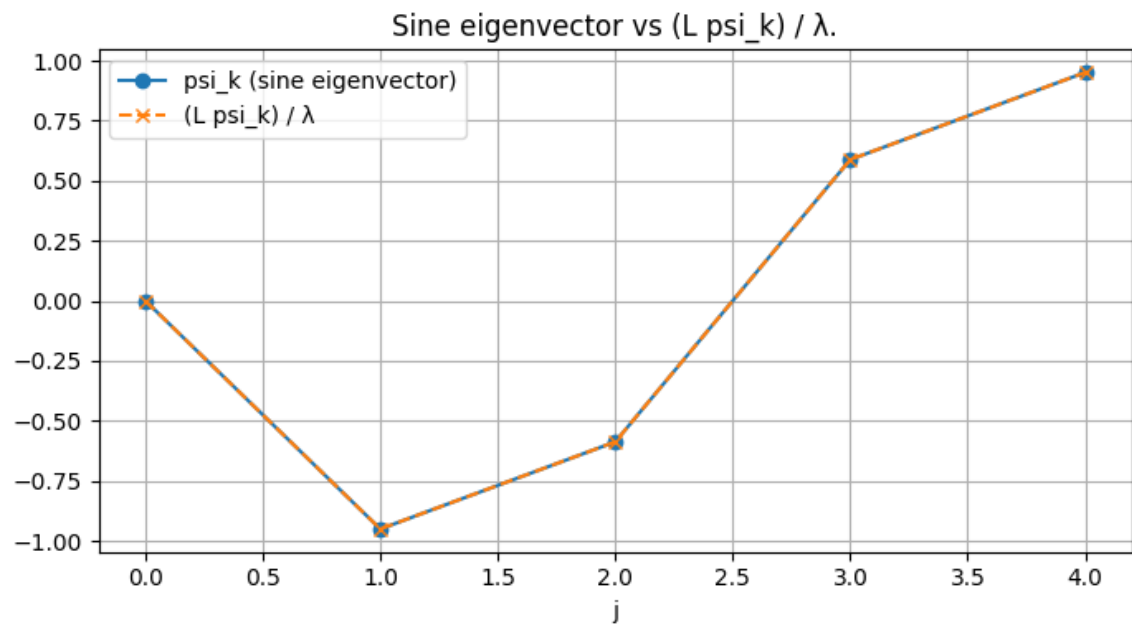
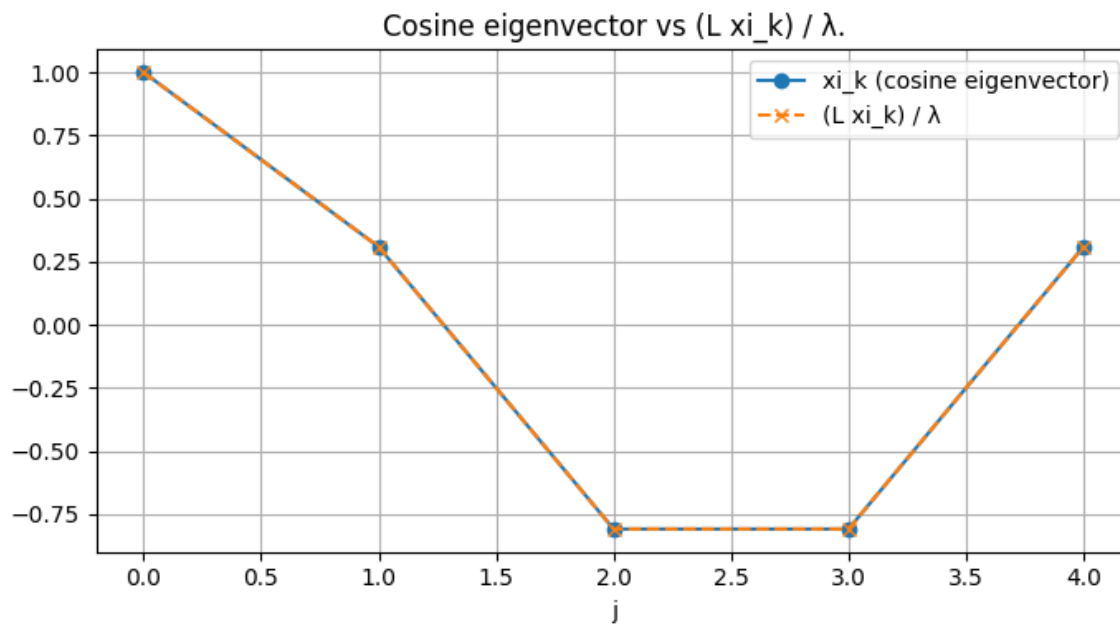
$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 & 1 \\ 1 & 0 & 1 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 0 & & 0 & 0 & 0 \\ & \vdots & & \ddots & & \vdots & \\ 0 & 0 & 0 & & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 1 & 0 & 1 \\ 1 & 0 & 0 & & 0 & 1 & 0 \end{bmatrix}$$

Eigenvectors:

$$\varphi_k = \left[\cos \frac{2\pi k}{n}, \cos \frac{4\pi k}{n}, \dots, \cos \frac{2(n-1)\pi k}{n} \right]$$

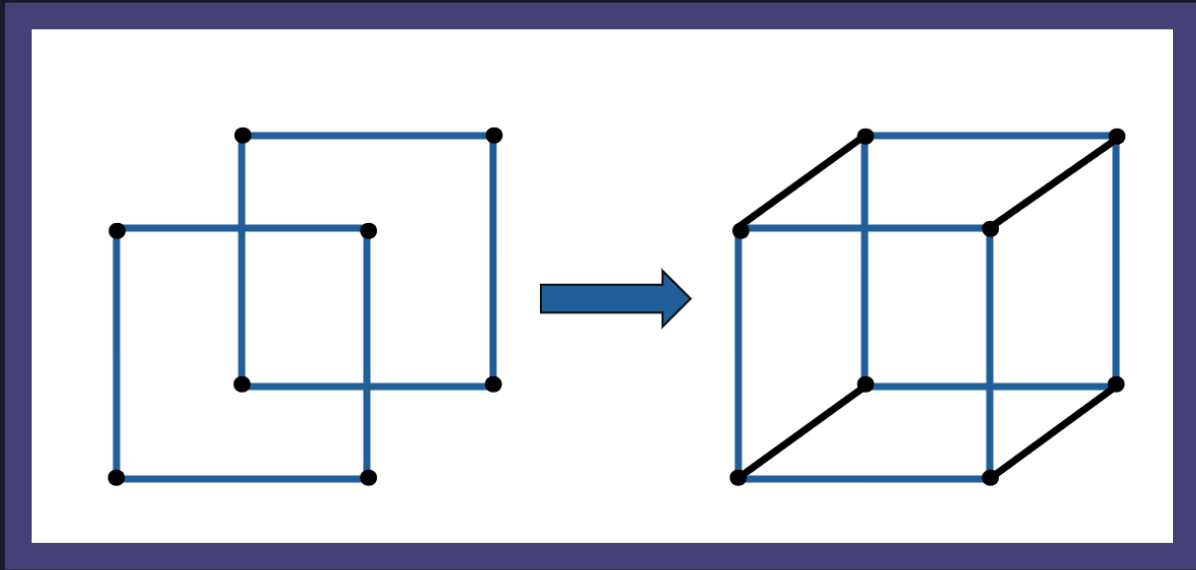
$$||L\xi - \lambda_k \xi||, ||L\psi - \lambda_k \psi|| = 0$$

Exercise 1 – Comparison



Exercise 2 – Spectrum of n -hypercube

Building the hypercube

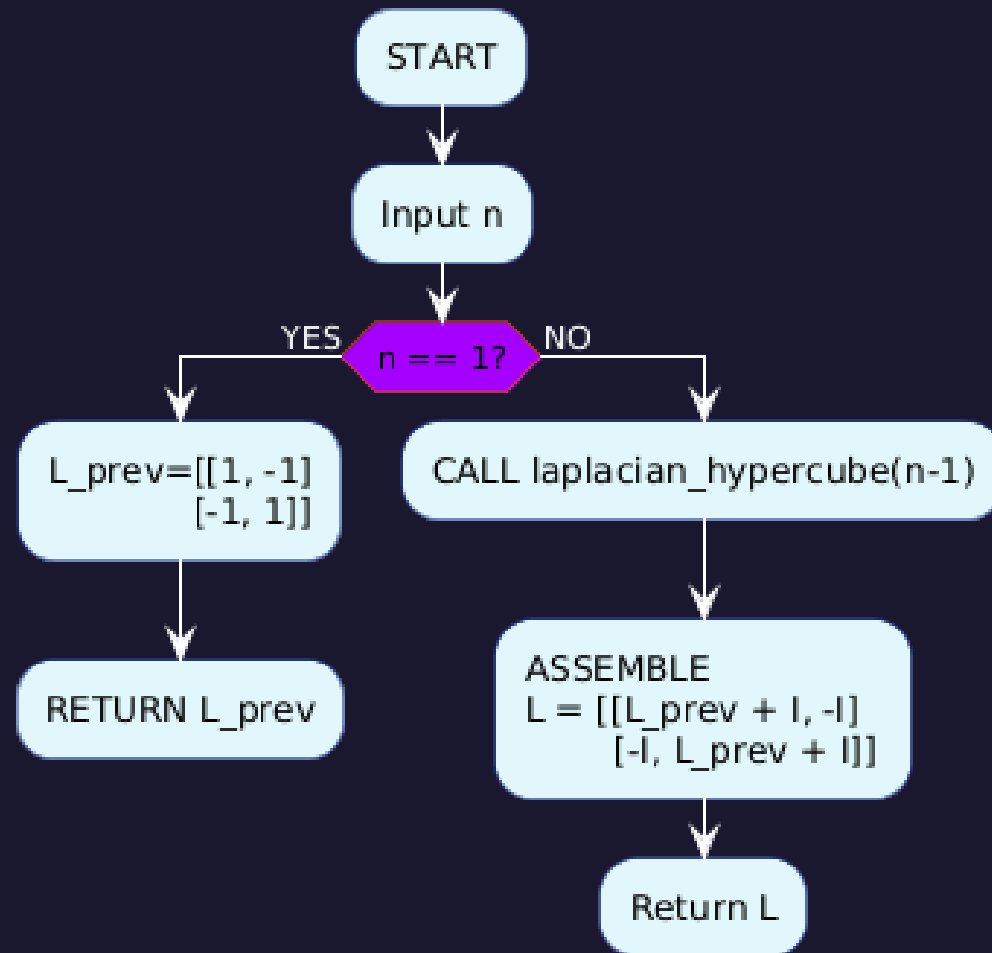


$$Q_{n+1} = \begin{bmatrix} Q_n & I \\ I & Q_n \end{bmatrix}$$



$$L(Q_{n+1}) = \begin{bmatrix} L(Q_n) + I & -I \\ -I & L(Q_n) + I \end{bmatrix}$$

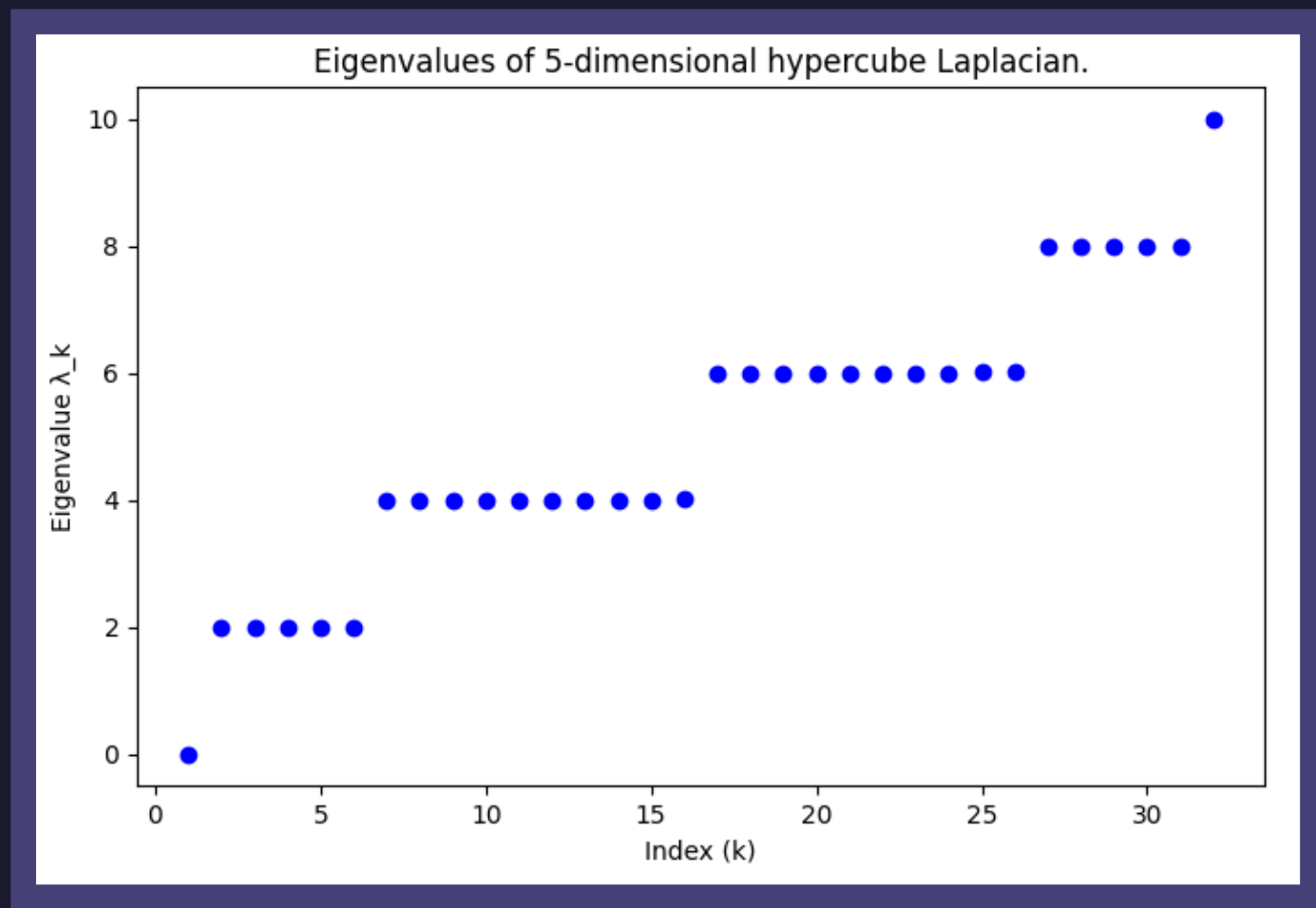
laplacian_hypercube workflow



Exercise 2 – Results

Spectrum: $\sigma(L(Q_n)) = \{0, n + 2, \dots, 2n\}$

Multiplicity: $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$



Thank you for
your attention

[Github Repository](#)

